



DR. B. C. ROY ENGINEERING COLLEGE, DURGAPUR

AN AUTONOMOUS INSTITUTE

AFFILIATED TO: MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST
BENGAL

Power System Planning & Reliability

PSM-104
M. Tech (EE)

Comprehensive Course Lesson Plan & Syllabus

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Academic Year 2026-27

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1 Course Overview

This course provides comprehensive coverage of power system planning methodologies and reliability assessment techniques. Students will develop expertise in short-term, medium-term, and long-term planning strategies while mastering reliability evaluation methods for generation, transmission, and distribution systems. The curriculum emphasizes practical application through case studies and computational tools.

-  **Course Fundamentals**
-  **Course Code:** PSM-104
-  **Credits:** 4
-  **Contact Hours:** 4 hours/week
-  **Department:** Electrical Engineering
-  **Institution:** Dr. B. C. Roy Engineering College, Durgapur

1.1 Prerequisites

-  Electric Power Systems
-  Electrical Measurements
-  Engineering Mathematics

2 Academic Calendar Alignment (Odd Semester 2026-27)

This lesson plan is aligned with the institution's *Provisional Academic Calendar for Academic Session 2026-27* (Odd Semester, applicable for students enrolled under Autonomy). All internal-assessment and examination windows below are reproduced from that calendar; course-specific dates (Assignment/Class Test) are fixed within those windows by the course instructor.

2.0.1 Institutional Dates (as per BCREC Academic Calendar 2026-27)

2.0.2 Course-Specific Assessment Dates (PSM-104)

 **Note:** Institutional dates are provisional and may be revised by the College as per prevailing circumstances; refer to the latest notice from the Controller of Examinations, BCREC. Course-specific dates (Assignment 2 / Class Test 2) will be confirmed and notified to students at least one week in advance.

3 Learning Objectives

Upon successful completion of this course, students will be able to:

Activity	Start Date	End Date
▶ Commencement of Academic Programme	03.08.2026	
✍ Submission of CIA1 marks (assessed & displayed before 16.09.2026)	16.09.2026	17.09.2026
✍ Submission of PCIA marks (assessed & displayed before 05.11.2026)	05.11.2026	06.11.2026
✍ Submission of CIA2 marks (assessed & displayed before 18.11.2026)	18.11.2026	19.11.2026
📝 Pre-Examination Activities (Exam form fill-up & verification)	23.11.2026	25.11.2026
📖 Theory/Practical/Sessional Examinations (Tentative)	30.11.2026	24.12.2026
📤 MAR Upload Window (Odd Semester)	03.08.2026	24.12.2026

Table 1: Key institutional dates, Odd Semester 2026-27 (Source: BCREC Provisional Academic Calendar 2026-27)

Activity	Date
✍ Assignment 1 (Modules 1 & 2, Option A/B): Submission	05.09.2026
✍ Class Test 1 / CIA01 (Modules 1 & 2)	09.09.2026
✍ CIA1 marks assessment & display (course-level)	On/before 16.09.2026
✍ Assignment 2 (Modules 3 & 5, Option A/B): Submission	On/before 05.11.2026 (PCIA window)
✍ Class Test 2 / CIA02 (Modules 3 to 5)	On/before 18.11.2026 (CIA2 window)

Table 2: Course-specific Assignment and Class Test dates, mapped to the institutional CIA windows above

Course Outcomes	Details/Statement	Action Verb	Knowledge Level
CO1	Analyze power system planning requirements across different time horizons	Analyze	Understand, Analyze
CO2	Evaluate system reliability using advanced probabilistic methods	Evaluate	Understand, Apply, Analyze
CO3	Design optimal generation expansion and transmission planning strategies	Design	Apply, Analyze, Create
CO4	Implement distribution system reliability enhancement techniques	Implement	Understand, Apply
CO5	Apply modern computational tools for power system analysis	Apply	Apply, Analyze

4 ≡ Course Structure

4.1 📅 Module 1: System Planning Fundamentals (7 Lectures)

🚩 Learning Outcomes:

- Understand the hierarchical nature of power system planning
- Identify key factors influencing planning decisions
- Distinguish between different planning horizons

📅 Week 1-2: Introduction to Power System Planning

Lecture	Topic	Duration	Learning Activities
1.1	Introduction to System Planning	50 min	🗣️ Interactive case study discussion
1.2	Planning Objectives and Constraints	50 min	🔧 Problem-solving workshop
1.3	Short-term Planning Strategies	50 min	📈 Load forecasting exercise
1.4	Medium-term Planning Approaches	50 min	💻 Capacity planning simulation
1.5	Long-term Planning Methodologies	50 min	📁 Strategic planning case study
1.6	Reactive Power Planning	50 min	💻 Octave/Scilab simulation concepts
1.7	Planning Assessment and Review	50 min	✅ Module assessment

🔑 Key Topics:

- Planning hierarchy and time scales
- Load forecasting techniques
- Economic and technical constraints
- Environmental considerations
- Regulatory framework impact

📌 **Module 1 Assessment:** Class Discussion on Planning case studies. Contributes to Assignment 1 (Modules 1 & 2), due 05.09.2026, and Class Test 1 / CIA01 (Modules 1 & 2), scheduled 09.09.2026.

4.2 Module 2: Reliability Theory and Methods (8 Lectures)

Learning Outcomes:

- Master probability concepts in reliability engineering
- Apply Markov processes to reliability problems
- Implement recursive techniques for reliability evaluation

Week 3-4: Probabilistic Reliability Analysis

Lecture	Topic	Duration	Learning Activities
2.1	Reliability Fundamentals	50 min	 Conceptual framework development
2.2	Failure Analysis and Probability Theory	50 min	 Statistical data analysis
2.3	Markov Process Applications	50 min	 State transition modeling
2.4	Recursive Reliability Techniques	50 min	 Algorithm implementation
2.5	Frequency and Duration Analysis	50 min	 Computational workshop
2.6	Stochastic Interruption Prediction	50 min	 Monte Carlo concepts
2.7	Reliability Cost Analysis	50 min	 Economic evaluation methods
2.8	Adequacy Assessment	50 min	 Comprehensive problem solving

Key Topics:

- Probability density functions
- Failure rate calculations
- Mean time to failure (MTTF)
- Availability and unavailability
- Cost-benefit analysis

 **Module 2 Assessment:** Class Discussion on Probability applications. Contributes to Assignment 1 (Modules 1 & 2), due 05.09.2026, and Class Test 1 / CIA01 (Modules 1 & 2), scheduled 09.09.2026.

4.3 Module 3: Generation Planning and Reliability (8 Lectures)

Learning Outcomes:

- Evaluate generation system reliability
- Optimize capacity expansion planning
- Assess interconnected system benefits

Week 5-6: Generation System Analysis

Lecture	Topic	Duration	Learning Activities
3.1	Generation Planning Overview	50 min	 Resource assessment workshop
3.2	Integrated Resource Planning	50 min	 Multi-criteria decision analysis
3.3	Generation System Modeling	50 min	 Stochastic modeling techniques
3.4	Loss of Load Probability (LOLP)	50 min	 Reliability index calculation
3.5	Capacity Expansion Strategies	50 min	 Optimization problem solving
3.6	Scheduled and Forced Outages	50 min	 Maintenance planning analysis
3.7	Interconnected System Analysis	50 min	 Multi-area system modeling
3.8	Emergency Assistance Protocols	50 min	 Contingency response planning

Key Topics:

- Generation mix optimization
- Load duration curves
- Capacity credit assessment
- Spinning reserve requirements
- Renewable energy integration

 **Module 3 Assessment:** Problem-solving session on Generation planning concepts, Programming Assignment on LOLP calculation and capacity optimization algorithms. Feeds into Assignment 2 (Modules 3 & 5) and the PCIA cycle (marks window 05.11.2026 to 06.11.2026).

4.4 Module 4: Transmission Planning and Reliability (7 Lectures)

Learning Outcomes:

- Design transmission networks for reliability
- Analyze composite system reliability
- Optimize network configurations

Week 7-8: Transmission System Design

Lecture	Topic	Duration	Learning Activities
4.1	Transmission Planning Objectives	50 min	 Network topology analysis
4.2	Network Reconfiguration Strategies	50 min	 Graph theory applications
4.3	System Reliability Indices	50 min	 Index calculation workshop
4.4	Load Point Reliability Analysis	50 min	 Nodal reliability assessment
4.5	Composite System Modeling	50 min	 Integrated G&T analysis
4.6	Data Requirements and Management	50 min	 Database design principles
4.7	Transmission Reliability Assessment	50 min	 Comprehensive case study

Key Topics:

- N-1 security criterion
- Transmission capacity planning
- Congestion management
- Power transfer capability
- Voltage stability considerations

 **Module 4 Assessment:** Class Discussion on Network topology analysis, Programming Assignment on Transmission reliability index calculations. Consolidation content for Class Test 2 / CIA02 (Modules 3 to 5).

4.5 Module 5: Distribution Planning and Reliability (10 Lectures)

Learning Outcomes:

- Analyze distribution system configurations
- Evaluate protection system impacts
- Optimize distribution reliability

📅 Week 9-11: Distribution System Analysis

Lecture	Topic	Duration	Learning Activities
5.1	Distribution System Fundamentals	50 min	🔌 Network topology review
5.2	Radial Network Analysis	50 min	📊 Reliability calculation methods
5.3	Network Reconfiguration Techniques	50 min	⚙️ Optimization algorithms
5.4	Interruption Indices (SAIFI, SAIDI)	50 min	% Index calculation practice
5.5	Protection System Impact Analysis	50 min	🔒 Protective device modeling
5.6	Disconnect and Transfer Effects	50 min	🔄 Switching strategy analysis
5.7	Distribution Reliability Indices	50 min	📈 Performance benchmarking
5.8	Parallel and Meshed Networks	50 min	🔌 Complex network analysis
5.9	Maintenance and Failure Analysis	50 min	✂️ Asset management strategies
5.10	Advanced Distribution Concepts	50 min	🏠 Smart grid integration

🔑 Key Topics:

- Feeder automation
- Distributed generation impact
- Load transfer capabilities
- Customer interruption costs
- Reliability-centered maintenance

📌 **Module 5 Assessment:** Class Test 2 / CIA02 (Modules 3 to 5) on Generation, Transmission and Distribution system concepts, scheduled on/before the CIA2 marks window (18.11.2026 to 19.11.2026). Programming Assignment on Distribution reliability analysis and SAIFI/SAIDI calculations, forming Assignment 2 (Modules 3 & 5).

5 📅 Assessment Framework

5.1 % Grading Distribution

📊 **Total Marks = End Semester Exam (60%) + Average of Assessments (40%)**

Component	Weight	Description
📅 End Semester Exam	60%	Comprehensive final examination
🏆 Higher Assessment	40%	Better of two assessments taken

Table 8: Assessment Component Weights

Assessment 1	Marks	Assessment 2	Marks
✍️ Class Test 1	25	✍️ Class Test 2	25
📄 Assignment 1	10	📄 Assignment 2	10
📋 Attendance	5	📋 Attendance	5
Total	40	Total	40

Table 9: Individual Assessment Breakdown

5.2 📋 Assessment Structure

Each assessment is evaluated out of 40 marks with the following distribution:

📌 **Assessment Selection:** The assessment with higher total marks is considered for final grade calculation. For example, if Assessment 1 = 37/40 and Assessment 2 = 39/40, then average of these Assessments (i.e. 38 marks) will be taken for final grading.

📋 CIA Mapping (Odd Semester 2026-27):

- **Assessment 1 = CIA01:** Class Test 1 (Modules 1 & 2) on 09.09.2026 + Assignment 1 (Modules 1 & 2, Option A or Option B, student's choice) due 05.09.2026 + Attendance. Marks assessed, displayed and submitted per the institutional CIA1 window (on/before 16.09.2026, submission 16.09.2026 to 17.09.2026).
- **PCIA (Institutional Checkpoint):** An additional college-level internal-assessment checkpoint per the Academic Calendar, with marks window 05.11.2026 to 06.11.2026. Module 3 content forms the basis of this checkpoint.
- **Assessment 2 = CIA02:** Class Test 2 (Modules 3 to 5) + Assignment 2 (Modules 3 & 5, Option A or Option B, student's choice) + Attendance, scheduled on/before the institutional CIA2 window (on/before 18.11.2026, submission 18.11.2026 to 19.11.2026).

5.3 🎓 Grading Scale

Grade	Range	Description
O	90-100	🏆 Outstanding
E	80-90	★ Excellent
A	70-80	👍 Very Good
B	60-70	✓ Good
C	50-60	— Average
D	40-50	👍 Pass
F	<40	✗ Fail

Table 10: Grade Distribution and Criteria

6 Learning Resources

6.1 Required Textbooks

1. **R.L. Sullivan**, "Power System Planning", Tata McGraw Hill Publishing Company Ltd.
2. **Roy Billinton & Ronald N. Allan**, "Reliability Evaluation of Power System", Springer Publication.

6.2 Supplementary References

3. **T.W. Berrie**, "Electricity Economics & Planning", Peter Peregrinus Ltd., London.
4. **IEEE Standards**: IEEE 493, IEEE 1366, IEEE 2030
5. **Research Papers**: Current publications from IEEE Transactions on Power Systems

6.3 Software Tools

- **GNU Octave / Scilab**: Open-source platforms for power system modeling and algorithm development
- **Python**: Custom reliability algorithms and data analysis
- **R**: Statistical analysis for reliability data and advanced analytics
- **Excel**: Data organization and basic computational analysis

7 Assignments (Home-based, Option A or Option B)

Each assignment carries 10 marks and offers a choice of two equally-weighted tracks. Students attempt **either** Option A or Option B, not both.

7.1 Assignment 1 (Modules 1 & 2): Load Forecasting & Two-State Markov Reliability

Format: Handwritten/typed solution sheet with complete derivations and working. **Submission**: 05.09.2026

Option A (Classical, Numerical): 10 Marks

Q1. (3 marks) Explain the weighted-sum approach to multi-objective power system planning, $\min\{w_1 f_{cost} + w_2 f_{reliability} + w_3 f_{environment}\}$. State what the weights w_i represent and why real-world planning solutions are balanced compromises rather than single-objective minima.

Q2. (3 marks) A utility has a current peak demand of 6,200 MW, an annual load growth rate of 4.0%, a planning reserve margin of 18%, and transmission losses of 3.5%. Determine (i) the projected peak demand for Year 4 of the planning horizon, and (ii) the required installed capacity for that year.

Q3. (2 marks) Three components connected in series have reliabilities 0.98, 0.95 and 0.97 respectively. Determine the overall system reliability.

Q4. (2 marks) A circuit breaker has a failure rate $\lambda = 0.08$ failures/year and a repair rate $\mu = 120$ repairs/year. Calculate its MTTF, MTTR (in hours) and steady-state availability.

🔗 Option B (Open-Source Coding): 10 Marks

Write a program in **Python, GNU Octave, or Scilab** (any one open-source platform) that:

(a) [5 marks] Implements the multiple linear regression load forecasting model $P_{L,t} = \beta_0 + \beta_1 T_t + \beta_2 H_t + \beta_3 D_t + \beta_4 E_t + \epsilon_t$ on a provided (or self-generated synthetic) 90-day dataset of temperature, humidity, day-type and load values. Report the fitted coefficients and the forecast load for a specified test day.

(b) [5 marks] Implements a two-state Markov reliability calculator that accepts failure rate λ and repair rate μ as inputs and returns MTTF, MTTR, steady-state availability A and unavailability U . Validate the routine against the two numerical cases in Assignment 1, Option A (Q4 above, and any worked example from the class notes).

Deliverables: Source code, a short analysis report, and validation output. **Tools:** Python / GNU Octave / Scilab (no proprietary software required).

7.2 🛡️ Assignment 2 (Modules 3 & 5): Generation Adequacy & Distribution Reliability

Format: Handwritten/typed solution sheet with complete derivations and working. **Submission:** On/before 05.11.2026 (PCIA window)

📋 Option A (Classical, Numerical): 10 Marks

Q1. (3 marks) A generation system has three identical units, each of capacity 120 MW and availability 0.90. (a) Using the recursive COPT construction method (Module 3, Lecture 3.3), build the Capacity Outage Probability Table for the system. (b) If the system peak load is 300 MW, calculate the Loss of Load Probability (LOLP) and the Loss of Load Expectation (LOLE) in days/year, and state whether the system meets the 0.1 days/year planning criterion.

Q2. (3 marks) A distribution system serves 10,000 customers through three feeders (component-based method, Module 5, Interruption Indices):

Feeder	Customers	λ (fail/yr)	r (hours)
X	2,800	1.0	3.0
Y	3,200	1.4	2.2
Z	4,000	0.9	2.8

Calculate SAIFI, SAIDI and CAIDI for the system.

Q3. (2 marks) A 250 MW solar PV plant is added to a system with a peak demand of 1,500 MW. Using a capacity credit of 0.22 (within the 10-70% range given for Solar PV in the Module 3 Capacity Credit table), determine (i) the effective firm capacity contributed by the plant, and (ii) the additional firm capacity still required to meet the peak demand.

Q4. (2 marks) A radial distribution feeder supplies a load point through two line sections in series (Module 5, Radial Network Reliability Model): Section 1 has $\lambda = 0.5$ failures/year, $r = 5$ hours; Section 2 has $\lambda = 0.7$ failures/year, $r = 4$ hours. Determine the load-point failure rate λ_{LP} , average interruption duration r_{LP} , and annual outage duration U_{LP} .

🔗 Option B (Open-Source Coding): 10 Marks

Write a program in **Python, GNU Octave, or Scilab** (any one open-source platform) that:

(a) [5 marks] Implements the three functions specified in the Module 3 Programming Assignment (Part 1): `build_copt_recursive(units)` using the recursive formula $P_n(C) = P_{n-1}(C) \times U_n + P_{n-1}(C - C_n) \times A_n$, `calculate_lolp(copt, peak_load)`, and `calculate_eue(copt, load_levels, durations)`. Validate against Assignment 2, Option A, Q1.

(b) [5 marks] Implements a distribution reliability calculator that computes SAIFI, SAIDI and CAIDI from a feeder-level input table (customers, λ , r per feeder), using the component-based formulas from Module 5. Validate against Assignment 2, Option A, Q2.

Deliverables: Source code, case study analysis, and validation output. **Tools:** Python / GNU Octave / Scilab. **Submission:** On/before 05.11.2026 (PCIA window), consolidated by the CIA2 window (18.11.2026 to 19.11.2026).

8 🗝️ Course Policies

8.1 📅 Attendance Policy

- Minimum 75% attendance required
- Medical leaves require proper documentation
- Make-up sessions available for excused absences

8.2 🕒 Late Assignment Policy

- 10% penalty per day for late submissions
- No submissions accepted after one week
- Extensions granted for documented emergencies

8.3 🎓 Academic Integrity

- All work must be original
- Proper citation required for references
- Collaboration encouraged but individual submission required

Week	Calendar Dates	Module	Topics	Assessments
1-2	03.08.2026 to 16.08.2026	1	System Planning Fundamentals	Assignment 1 (Mod. 1 & 2) in progress
3-4	17.08.2026 to 30.08.2026	2	Reliability Theory	Assignment 1 due 05.09.2026;  Class Test 1 / CIA01 on 09.09.2026
5-6	31.08.2026 to 13.09.2026	3	Generation Planning	CIA1 marks window 16.09.2026 to 17.09.2026
7-8	14.09.2026 to 27.09.2026	4	Transmission Planning	 Class Discussion
9-11	28.09.2026 to 18.10.2026	5	Distribution Planning	Assignment 2 (Mod. 3 & 5) in progress
Consolidation	19.10.2026 to 22.11.2026	3-5	Revision & doubt-clearing	PCIA window 05.11.2026 to 06.11.2026;  Class Test 2 / CIA02 and Assignment 2 due on/before 18.11.2026; CIA2 window 18.11.2026 to 19.11.2026
Pre-Exam	23.11.2026 to 25.11.2026	Exam form fill-up & verification		 Pre-Examination Activities
Exam	30.11.2026 to 24.12.2026	End Semester Examination		 Theory/Practical/Sessional (Tentative)

Table 11: Semester Overview and Assessment Schedule (Odd Semester 2026-27, per BCREC Academic Calendar)

9 Course Schedule Summary

10 Faculty Contact Information

 **Course Instructor:** Kingsuk Majumdar, Ph.D (Engg)
 **Department:** Electrical Engineering
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 **Office Hours:** Monday-Friday, 9:45 AM to 5:45 PM
 **Contact:** Available during office hours for consultation